CYBER SECURITY

UNIT 1 – COMPLETE DETAILED NOTES

Cyber Crime • Challenges of Cyber Crime • Classifications of Cyber Crimes

Email Spoofing | Spamming | Internet Time Theft | Salami Attack

Book-style notes with Practical Examples, Definitions, Flowcharts, Diagrams, Important Points, Exam Notes & Memory Tricks

UNIT 1 – SYLLABUS (As per Examination)

How to read this book: Har topic me sabse pehle ek **long practical example** diya hai, phir **short definition**, phir **flowchart / diagram**, phir **important points** aur aakhir me **exam notes**. Isi order me padhna — definition yaad aasan ho jayegi.

CONTENTS

PART 1 – CYBER SECURITY (Complete)

Unit 1 ki pehli building block. Cyber Security samajhne ke liye pehle ye janna zaroori hai ki wo kya hoti hai, kisko protect karti hai, aur kyu zaroori hai. Aage ke saare chapters (Cyber Attack, Cyber Crime) isi par khade hain.

1.1 Cyber Security Kya Hoti Hai?

PRACTICAL EXAMPLE – REAL LIFE

Socho tumhare Papa ne ek naya ghar banaya. Us ghar me bahut important cheezein hain: Rs. 5 lakh cash, gold jewellery, laptop, documents (Aadhaar, PAN, Property Papers), TV, mobile, family photos.
Ab tumhare Papa **bina lock lagaye** ghar chhod dete hain. Kya ho sakta hai? Chor andar aa sakta hai, paise chori ho sakte hain, jewellery le ja sakta hai, documents jala sakta hai, ghar ka samaan tod sakta hai.
Isliye Papa security lagate hain: **strong door, lock, CCTV, alarm, security guard, fingerprint lock**. Ab chor ke liye ghar me ghusna bahut mushkil ho gaya.

PRACTICAL EXAMPLE – REAL LIFE

Aaj har insaan ka ek **Digital Ghar** hota hai. Usme kya hota hai? Gmail, WhatsApp, Instagram, Facebook, College Portal, Bank Account, UPI, ATM Details, Aadhaar, PAN, Photos, Passwords, Company Files.
Ab socho agar koi hacker is digital ghar me ghus jaye to wo kya kar sakta hai? Password chori, bank account se paise transfer, photos leak, Gmail hack, Instagram account delete, college marks change, company ka data chori.
Bilkul waise hi jaise chor tumhare ghar me ghus gaya tha. Is digital ghar ko bachane ke liye jo security lagayi jaati hai, usse **Cyber Security** kehte hain.

DEFINITION

Cyber Security woh process hai jo computers, mobiles, servers, networks aur data ko hackers, viruses aur unauthorized access se protect karti hai.

FLOWCHART

```
Computer / Mobile
          │
          ▼
Important Data
          │
          ▼
Cyber Security
          │
┌─────────┬──────────┐
│         │           │
Hackers   Virus   Data Theft
Blocked   Blocked   Prevented
```

1.2 Cyber Security Kiski Security Karti Hai?

Cyber Security **sirf computer ki security nahi karti**. Ye protect karti hai:

- Computer, Mobile, Network, Cloud, Database, Server
- Email, Online Banking, Online Shopping
- Hospital Records, College Data, Company Information

1.3 Cyber Security Ki Zarurat Kyu Hai?

Agar Cyber Security na ho to:

- Password chori ho sakta hai.

- ATM account hack ho sakta hai, WhatsApp hack ho sakta hai.
- Company ka data leak ho sakta hai.
- Hospital records delete ho sakte hain.
- Government websites hack ho sakti hain.

Daily Life Examples: ATM PIN, Phone Lock, Face Unlock, OTP, Gmail Password, Antivirus, Firewall — ye sab choti-choti security hi to hain jo hamari digital life protect karti hain.

1.4 Cyber Security Ke Objectives (Goals)

- **Data ko safe rakhna** — jaise bank balance.
- **Privacy bachana** — jaise WhatsApp chats.
- **Unauthorized access rokna** — sirf owner hi account khol sake.
- **Virus rokna** — computer ko damage hone se bachana.
- **Online fraud rokna** — fake websites aur scams se protection.

1.5 Cyber Security Ke Main Components

FLOWCHART

```
Cyber Security
       |
┌──────┬─────────┐
|         |            |
Network   Application    Information
Security   Security      Security
```

Aur bhi components hote hain: **Cloud Security, Mobile Security, IoT Security, Endpoint Security**.

1.6 Cyber Security Kahan Use Hoti Hai?

Jagah	Kyu
Bank	Money protect karne ke liye
Hospital	Patient records protect karne ke liye
College	Student data aur marks protect karne ke liye
Amazon / Flipkart	Customer payment secure karne ke liye
Airport	Flight systems ko secure rakhne ke liye
Army	Secret military information protect karne ke liye

1.7 Cyber Security Ke Benefits

- Data Safe, Privacy Safe, Money Safe, Identity Safe
- Business Safe, Trust Increase
- Online Transactions Safe

1.8 Agar Cyber Security Na Ho To?

Imagine ek din: tumhara Gmail hack, WhatsApp delete, Instagram kisi aur ke control me, bank account se Rs. 50,000 transfer, college portal me marks change, laptop ki files encrypt. **Ye sab ek hi din me ho sakta hai** — isi liye Cyber Security bahut important hai.

EXAM NOTES (2 MARKS / VIVA)

Exam Definition (2 Marks): Cyber Security is the practice of protecting computers, networks, devices and data from unauthorized access, cyber attacks and digital threats.

Viva Q1. Cyber Security kya protect karti hai?
Ans: Computer, Network, Server, Mobile aur Data.

Viva Q2. Cyber Security ka main purpose kya hai?
Ans: Data ko unauthorized access aur attacks se protect karna.

MEMORY TRICK

Cyber Security = Digital Ghar ki Security. Jaise ghar ko lock, CCTV aur guard protect karte hain, waise hi Cyber Security hamare digital data ko hackers aur attacks se protect karti hai.

PART 2 – CYBER ATTACK (Complete)

Foundation chapter. Cyber Attack ka concept samajhna zaroori hai kyunki Cyber Crime (Unit 1) aur Active/Passive Attack dono isi ki branches hain. Dhyan se padho — ye sabki root hai.

2.1 Cyber Attack Kya Hota Hai?

PRACTICAL EXAMPLE – REAL LIFE

Socho tumhare school me ek **Computer Lab** hai jisme 40 computers hain. Usme save hai: students ki marksheet, fees records, teachers ki salary, exam papers, principal ke documents, CCTV recordings.
Ab maan lo ek din ek hacker internet ke through school ke server se connect ho gaya. Wo kya kar sakta hai?
Situation 1: Sabhi students ke marks change kar deta hai — 90 → 40, 40 → 95.
Situation 2: Principal ka password change kar deta hai — ab Principal khud login nahi kar pa rahe.
Situation 3: Exam paper internet par leak kar deta hai.
Situation 4: Saari files delete kar deta hai.
Situation 5: School ki website band kar deta hai.
Situation 6: Students ki personal information (mobile, Aadhaar, address, parents details) chori kar leta hai.
Socho, school ko kitna bada loss hoga. Ye jo hacker ne **jaan-boojhkar** system ko nuksan pahunchaya, ise **Cyber Attack** kehte hain.

PRACTICAL EXAMPLE – REAL LIFE

Tumhara Gmail password hai **Vinay@123**. Ek hacker ne kisi tarah password jaan liya. Fir usne Gmail login kiya, password change kar diya, recovery email badal di aur OTP number bhi badal diya.
Ab Gmail tumhara tha, lekin control hacker ke paas chala gaya. **Ye bhi Cyber Attack hai.**

DEFINITION

Cyber Attack ek intentional (jaan-boojhkar) attempt hota hai jisme attacker computer, network, server ya data ko access, steal, damage, modify ya destroy karne ki koshish karta hai.

FLOWCHART

```
Attacker
    |
    ▼
Internet
    |
    ▼
Victim System
    |
    ▼
Data Damage / Theft / Control
```

2.2 Cyber Attack Me 3 Main Characters Hote Hain

DIAGRAM

```
Cyber Attack
     |
 ┌────┬────┐
 |     |      |
Attacker Target  Result
```

- **Attacker** — jo attack karta hai: Hacker, Criminal, Terrorist, Insider Employee.
- **Target (Victim)** — jis par attack hota hai: Mobile, Laptop, Bank, Hospital, School, Company, Government Website.
- **Result** — attack ke baad kya hua: Data chori, Password leak, Website band, Virus install, Money transfer.

2.3 Cyber Attack Kyu Kiya Jata Hai? (Motives)

Bahut log sochte hain hacker sirf **maze ke liye** hack karta hai. Reality me reasons alag hote hain:

Motives	Example
1. Money	Bank account hack, UPI fraud, Credit Card fraud (sabse common)
2. Data Chori	Company ke secret documents chori karna
3. Identity Chori	Tumhare naam se fake account banana
4. Revenge	Ex-employee company ka database delete kar deta hai
5. Fame	Kuch hackers famous hone ke liye website hack karte hain
6. Political Reason	Ek country dusri country ke systems par attack karti hai
7. Terrorism	Power grid, airport ya railway system ko target karna

2.4 Cyber Attack Ka Process (Attack Lifecycle)

Har attack ek hi second me nahi hota. Generally ye steps hote hain:

FLOWCHART

```
Step 1: Information Collect
           │
           ▼
Step 2: Weakness Find
           │
           ▼
Step 3: Attack Launch
           │
           ▼
Step 4: Gain Access
           │
           ▼
Step 5: Steal / Modify / Destroy
```

- **Step 1 – Information Collection:** Hacker pehle victim ke baare me info collect karta hai — naam, email, mobile, website, IP address.
- **Step 2 – Weakness Find:** Dekhta hai system me vulnerability kahan hai — weak password, old software, no antivirus.
- **Step 3 – Attack Launch:** Ab hacker attack karta hai.
- **Step 4 – Access:** System ke andar ghus jata hai.
- **Step 5 – Damage:** Ab jo chahe karta hai — delete, copy, encrypt, leak.

2.5 Cyber Attack Kin Cheezon Par Ho Sakta Hai?

Computer, Mobile, Wi-Fi, Server, Website, Email, Cloud, ATM, Smart TV, CCTV, IoT Devices — **koi bhi digital device target ban sakta hai.**

2.6 Cyber Attack Ke Main Categories

FLOWCHART

```
Cyber Attack
      │
  ┌───┴────┐
  │         │
Active      Passive
Attack      Attack
```

Active Attack (jaise DoS, DDoS, Message Modification, Masquerade) aur **Passive Attack** (jaise Eavesdropping, Packet Sniffing, Traffic Analysis) — inhe aage ki books me detail me padha jata hai. Cyber Crime ke liye bas itna yaad rakho ki **attack hi crime ka jariya hota hai**.

APPENDIX C – CYBER SECURITY MASTER TREE

Cyber Security ka **poora map** ek hi diagram me — isme Cyber Attack ki types (Active/Passive) aur Malware ki types dono dikhti hain. Ye exam me diagram ke liye ekdum ready hai.

DIAGRAM – CYBER SECURITY MASTER TREE

```
Cyber Security
      │
      ├──────────────┐
      │              │
Cyber Attack     Malware
      │              │
      │              ├── Virus
      │              ├── Worm
      │              ├── Trojan
      │              ├── Spyware
      │              ├── Adware
      │              ├── Ransomware
      │              └── Rootkit
      │
      ├── Active Attack
      │       ├── DoS
      │       ├── DDoS
      │       ├── Message Modification
      │       ├── Fabrication
      │       └── Malware-based Attack
      │
      └── Passive Attack
              ├── Eavesdropping
              ├── Packet Sniffing
              ├── Traffic Analysis
              └── Wiretapping
```

Active Attack = jo system me **badlaav** karta hai (modify/destroy) — jaise DoS, DDoS, Message Modification, Fabrication, Malware-based Attack. **Passive Attack** = jo sirf **dekh-ta hai** (data churakar padhta hai, badlaav nahi karta) — jaise Eavesdropping, Packet Sniffing, Traffic Analysis, Wiretapping.

Attack	Type	Kya hota hai
DoS / DDoS	Active	System par overload karke service band karna
Message Modification	Active	Message ke andar data badal dena
Fabrication	Active	Jhootha / fake message ghusa dena
Malware-based Attack	Active	Virus/ransomware install kar ke nuksan
Eavesdropping	Passive	Chupke se baat-chit sunna
Packet Sniffing	Passive	Network traffic ke packets padhna
Traffic Analysis	Passive	Data ko padhe bina uske patterns se info nikalna
Wiretapping	Passive	Kisi aur ki line/communication tap karna

APPENDIX B – MALWARE TREE

Malware = **Mal**icious soft**ware** — wo programs jo computer ko nuksan pahunchate hain. Ye exact tree exam ke diagram sawal ke liye ready hai.

DIAGRAM – MALWARE CLASSIFICATION TREE

```
Malware
    │
  ┌──┬────────────────────────────────────┐
  │  │   │       │       │       │
Virus Worm Trojan Spyware Adware Ransomware
                              │
                    Rootkit
                    Keylogger
                    Bot
                    Botnet
```

Ye tree exam me 3 tarah se poochha jata hai: (1) Malware kya hai — definition, (2) Malware ki types — tree diagram, (3) kisi ek type ko explain karo — table se.

Malware	Kya karta hai	Example
Virus	Files me chhip kar phailta hai, damage karta hai	ILOVEYOU
Worm	Bina kisi file ke khud copy hoke network par phailta hai	Morris Worm
Trojan	Achha software ban ke andar ghus jata hai	Banking trojans
Spyware	Bina bataye aapki activity dekhta hai	Keyloggers, trackers
Adware	Zabardasti ads dikhata hai	Pop-up ads
Ransomware	Data lock (encrypt) karke paisa maangta hai	WannaCry
Rootkit	System ki jad me chhip jata hai, detection se bachta hai	Hiding in kernel
Keylogger	Har key press record karta hai (password churata hai)	Bank password steal
Bot / Botnet	Aapka system hacker ke remote control me	Spam-sending botnets

APPENDIX A – SECURITY SERVICES (CIA TRIAD)

Cyber Security ka poora foundation **CIA Triad** par khada hai. Exam me ye sabse poocha jaane wala diagram hai. Pehle **exact diagram**, phir explanation.

DIAGRAM – CIA TRIAD

```
CIA
 │
 ├── Confidentiality → Kaun dekh sakta hai?
 ├── Integrity       → Data badla ya nahi?
 └── Availability    → Zarurat par mil raha hai?
```

DIAGRAM – BAQI SERVICES

```
Baaki Security Services:

Authentication   → Tum kaun ho?
Authorization    → Tum kya kar sakte ho?
Accountability   → Kisne kya kiya?
Non-Repudiation  → Kiya hai to mana nahi kar sakte.
Accessibility    → Authorized user aasani se use kar sake.
Privacy          → Personal information ko protect karna.
```

DEFINITION

Confidentiality: Data ko sirf authorized (allowed) log dekh saken.
Integrity: Data ko koi badal / corrupt na kar sake.
Availability: Authorized user ko zarurat par data hamesha milta rahe.

Service	Sawaal	Matlab	Attack Jo Todta Hai
Confidentiality	Kaun dekh sakta hai?	Data sirf authorized log hi dekh saken	Eavesdropping, Sniffing
Integrity	Data badla ya nahi?	Data original aur galat nahi hona chahiye	Message Modification
Availability	Zarurat par mil raha hai?	Authorized user ko kabhi bhi data mile	DoS / DDoS

Service	Sawaal	Matlab
Authentication	Tum kaun ho?	Pehchan verify karna (password, OTP, fingerprint)
Authorization	Tum kya kar sakte ho?	Access level decide karna (kya allowed hai)
Accountability	Kisne kya kiya?	Har kaam ka record hona chahiye
Non-Repudiation	Kiya hai to mana nahi kar sakte	Kiya hua kaam deny nahi kar sakte (digital signature)
Accessibility	Aasani se use kar sake	System sirf aapke liye kaam kare
Privacy	Personal info protect	Personal data kisi aur ke haath na jaye

2.7 Cyber Attack Ke Possible Results

Result	Example
Data Theft	Password chori
Data Modification	Marks badal dena
Data Deletion	Files delete
Financial Loss	Bank fraud
Service Down	Website crash
Privacy Loss	Personal photos leak

2.8 Cyber Attack Ko Rokne Ke Tarike (Prevention)

- Strong Password, 2-Factor Authentication (2FA)
- Antivirus, Firewall, Software Update, Backup
- Unknown links par click na karna, Public Wi-Fi par sensitive kaam na karna

EXAM NOTES (2 MARKS / VIVA)

Exam Definition (2 Marks): Cyber Attack is an intentional attempt to gain unauthorized access to a computer system, network, or data in order to steal, modify, destroy, or disrupt information or services.

Viva Q1. Cyber Attack kya hai?
Ans: Kisi computer ya network par unauthorized access ya damage pahunchane ki koshish ko Cyber Attack kehte hain.

Viva Q2. Cyber Attack ka sabse common objective kya hai?
Ans: Data theft, financial gain, aur system ko damage karna.

MEMORY TRICK

Cyber Attack = Digital Chor. Jaise ghar me chor ghuskar paisa ya samaan chori karta hai, waise hi cyber attack me attacker digital system me ghuskar data, password ya paise chori karta hai ya system ko nuksan pahunchata hai.

PART 3 – INTRODUCTION OF CYBER CRIME

Unit 1 ka asli syllabus yahan se shuru hota hai. Is chapter me hum samjhenge ki Cyber Crime kya hai, iske elements kya hain, ye traditional crime se kaise alag hai, aur kis kis category me divide hota hai.

3.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Meera Delhi me ek bank me kaam karti hai. Ek din uske phone par WhatsApp message aaya:
"Congratulations! Aapne Rs. 5,00,000 ki lottery jeeti hai. Claim karne ke liye is link par click karein."
Meera ne link click kiya. Ek page khula jisme likha tha — **"Processing fee Rs. 2,000 bharo"**. Usne UPI se bhar diye. Phir aise hi do aur fees bhari. Total **Rs. 12,000** chale gaye. Lottery ka paisa kabhi nahi aaya. **Ye ek Cyber Crime hai (Online Fraud).**

Usi hafte aur bhi kuch hua:

- Meera ke friend **Rohit** ke Instagram account se uske followers ko "free iPhone" messages gaye — account hack ho gaya.
- College portal me kisi ne **marks change** kar diye.
- Ek kaam-wali didi ke **Aadhaar se fake SIM** nikal li gayi.

Ye saare incidents ek cheez me same hain — **computer / mobile / internet ka istemal karke criminal kaam hua.** Isi ko hum Cyber Crime kehte hain.

3.2 Definition

DEFINITION

Cyber Crime woh criminal activity hai jisme computer, mobile ya internet ka istemal karke koi illegal kaam kiya jata hai — jaise paisa chori karna, data chori karna, kisi ko dhamkana, ya kisi system ko nuksan pahunchana.

Simple English: Any criminal activity that involves a computer or the internet is called Cyber Crime.

3.3 Flowchart

FLOWCHART

```
Criminal (Offender)
      |
      ▼
Computer / Mobile / Internet  (Tool)
      |
      ▼
Victim (Person / Company / Govt.)
      |
      ▼
Loss (Money / Data / Privacy / Trust)
```

3.4 Elements of Cyber Crime (5 Cheezein Zaroori)

Kisi bhi cyber crime me ye 5 elements hote hain — agar ek bhi missing ho to cyber crime nahi banega:

Element	Kya hai	Example
1. Offender	Jo galat kaam karta hai	Hacker, fraud gang, insider employee
2. Tool	Jis se galat kaam hota hai	Computer, internet, mobile, software
3. Victim	Jis par galat kaam hota hai	Person, company, bank, government

4. Act	Illegal activity khud	Hacking, fraud, spoofing, spam
5. Loss / Impact	Hone wala nuksan	Money, data, privacy, reputation

3.5 Cyber Crime vs Traditional Crime

Cyber Crime ko samajhne ka sabse aasan tarika hai ise **normal (traditional) crime** se compare karna:

Basis	Traditional Crime	Cyber Crime
Jagah	Sirf ek jagah (local)	Internet hai to poore world me
Tool	Physical (chaku, bandook)	Computer, internet, software
Criminal ki pehchan	Pata chalti hai (face, CCTV)	Chhupi hoti hai (fake ID, VPN)
Evidence	Physical (fingerprint, weapon)	Digital (logs, IP, files) — delete ho sakta hai
Speed	Dheere hota hai	Seconds me khatam
Distance	Criminal ko paas hona zaroori	Duniya ke kisi bhi kone se
Jurisdiction	Ek hi desh/court ka law	Kaunsa desh ka law lagega — confusion
Record	Bahut kam records	Sab kuch digital record rehta hai

3.6 Classifications of Cyber Crimes (Overview)

Cyber Crimes ko target ke hisaab se **3 bade categories** me divide kiya jata hai:

FLOWCHART

```
Cyber Crime
      │
 ┌────┴──────────────────┐
 │                     │                        │
Against              Against                 Against Government
Persons              Property                 / Society
 │                     │                        │
 │   Harassment          │   Hacking               │   Cyber Terrorism
 │   Cyber Stalking      │   Phishing              │   Cyber Warfare
 │   Identity Theft      │   Email Spoofing        │   Website Defacement
 │                      │   Spamming
 │                      │   Internet Time Theft
 │                      │   Salami Attack
 │                      │   Software Piracy
```

IMPORTANT POINTS

Unit 1 syllabus me **4 classifications** detail me padhenge: **Email Spoofing, Spamming, Internet Time Theft, Salami Attack** — ye sab 'Against Property' category ke cyber crimes hain. Inhe Part 5 me poore detail me padha jayega.

3.7 Characteristics of Cyber Crime

- **Anonymity** — criminal apni pehchan chhupa kar kaam karta hai.
- **Global reach** — duniya ke kisi bhi country ko target kar sakta hai.
- **Low cost** — bas ek computer + internet chahiye.
- **No boundaries** — borders nahi maante.
- **Instant** — seconds me kaam ho jata hai.
- **Hard to detect** — evidence delete karna aasan hai.
- **High impact** — ek attack se lakhon logon ko nuksan ho sakta hai.

3.8 Who Commits Cyber Crime? (Types of Cyber Criminals)

Criminal Type	Kaun hai	Kya karta hai
Hackers	White / Black / Grey hat	System me ghusne wale technical log
Organized Criminals	Gangs	Bank fraud, card fraud, extortion
Insiders	Company ke andar ka employee	Andar ki information leak karta hai
Terrorists	Desh ke khilaf groups	Cyber terrorism, critical systems par attack
Disgruntled Employees	Ex-employee	Badla lene ke liye data delete/leak
Script Kiddies	Naye log	Ready-made tools use karke khelne wale

3.9 Real World Examples (Chote Chote)

- **WannaCry Ransomware (2017)** — duniya bhar ke hospitals aur laptops band kar diye.
- **Bank SIM-swap frauds** in India — SIM block karke OTP chura kar paise nikalna.
- **Aadhaar data leaks** — personal data market me bikna.
- **Deepfake videos** — kisi ki fake video banakar blackmail.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes – One Line Definitions:
1. Cyber Crime = computer/internet ke through ki gayi criminal activity.
2. Cyber Crime ke 5 elements = Offender, Tool, Victim, Act, Loss.
3. Cyber Crime ki 3 categories = Against Person, Against Property, Against Government.
4. Cyber crime me criminal chhupa rehta hai, par digital evidence rehta hai.

Viva Q1. Cyber Crime me 'tool' kya hota hai?
Ans: Computer, mobile, ya internet jiske through crime hota hai.

Viva Q2. Cyber crime ko traditional crime se alag kya banata hai?
Ans: Anonymity, global reach, speed aur digital evidence — criminal ko victim ke paas hone ki zaroorat nahi.

MEMORY TRICK

Cyber Crime = Digital Daku. Jaise highway par daku gaadi rok kar lootta hai, waise hi internet par cyber criminal system rok kar data aur paise lootta hai — bas farak itna hai ki daku ko face dikhata hai, cyber criminal nahi.

PART 4 – CHALLENGES OF CYBER CRIME

Crime hota hai, lekin usse pakadna itna aasan nahi hota. Is chapter me hum wo saari problems padhenge jo cyber crimes ko rokne, pakadne aur saza dilane me aati hain.

4.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ravi ke bank account se Rs. 2,00,000 gayab ho gaye. Police ne case khola. Lekin investigation me har step par rukawat aayi:

1. Account se paisa 3 alag accounts me gaya, phir 50 aur accounts me transfer hua.
2. Sab transactions **VPN** se hui — IP address fake tha.
3. Attacker South Africa me tha, server America me, victim India me.
4. Paisa cryptocurrency (Bitcoin) me badal diya gaya — track karna near impossible.
5. Jab tak police ko pata chala, 6 mahine nikal chuke the — digital records delete ho chuke the.

Police ke saamne problem: **criminal pakadna mushkil, saza dilana aur bhi mushkil.** Yehi problems — jo cyber crime se nipatne me aati hain — **Challenges of Cyber Crime** kehlati hain.

4.2 Definition

DEFINITION

Challenges of Cyber Crime wo difficulties / problems hain jo cyber crimes ko **prevent (rokne), detect (pakadne), investigate (jaanch) aur punish (saza dilane)** me saamne aati hain.

Simple: Crime karna aasan, lekin cyber crime ko pakadna — kyunki criminal chhupa hua hai, country ke bahar hai, aur evidence mit jata hai.

4.3 Flowchart – Challenges Ki Chain

FLOWCHART

```
Criminal commits cyber crime
        │
        ▼
Crime is fast & automated
        │
        ▼
No borders / wrong jurisdiction
        │
        ▼
Identity hidden (VPN, fake ID, crypto)
        │
        ▼
Evidence volatile / easy to delete
        │
        ▼
Investigation slow & costly
        │
        ▼
Lack of laws & awareness
        │
        ▼
Very few criminals get punished
```

4.4 Detailed Challenges (Sabhi Point Detail Me)

1. Anonymity (Pehchan Chhupana)

Attacker **VPN, Tor browser, fake accounts aur stolen SIM** istemal karta hai jisse uski asli pehchan kabhi saamne nahi aati. **Example:** fraud karne ke baad account fake naam se khula milta hai, asli banda kisi aur country me hota hai.

2. Jurisdiction Problem (Kaunsi Police Case Kare)

Crime **India** me hua, server **USA** me hai, attacker **Russia** me baitha hai. Ab kaunsi country ki police case kare? Har desh ke laws alag hote hain, aur cross-border cooperation me mahino lag jaate hain.

3. Global / Borderless Nature

Internet par koi border nahi hota. Ek click me attacker duniya ke kisi bhi system par attack kar sakta hai. **Example:** ek country ke hacker dusre country ke bank par hamla — bina wahan gaye.

4. Evidence Collection (Saboot Jutana Mushkil)

Digital evidence (logs, cookies, IP records, files) **bahut nazuk** hota hai — computer band karo ya system restart karo to data jaa sakta hai. Delete karna bhi ek click ka kaam hai.

5. Fast & Automated Attacks

Bots aur scripts ek second me lakhon attacks kar dete hain. Police/security ke paas itna time hi nahi hota ki wo har attack ko rok saken.

6. Encryption (Data Lock)

Attacker apna data **encrypt** kar leta hai (jise bina key ke koi nahi padh sakta). Police ke liye encrypted data ko padhna aur prove karna bahut mushkil ho jata hai.

7. Rapidly Changing Technology

Roz nayi technology aur naye tareeke ke crimes bante hain. **Laws aur police piche reh jaate hain** kyunki naya crime samajhne me time lagta hai.

8. Lack of Awareness (Sabse Bada Weak Point)

Zyaada tar log OTP share kar dete hain, unknown links par click karte hain, weak passwords rakhte hain. **Insaan hi sabse bada weak link hai.** Cyber security itni strong nahi hoti ki user ko bacha sake.

9. Lack of Technical Skills in Investigation

Aam police ko cyber ka training nahi hoti. **Cyber cell / forensic lab mahangi aur kam** hoti hai, isliye investigation slow aur incomplete rehti hai.

10. Cross-border Legal Differences

Kuch desh me cyber crime ke **strict laws hi nahi hain** — aise desh cyber criminals ke liye safe haven ban jaate hain jahan se wo bina dar ke kaam karte hain.

11. High Cost of Security

Strong security (firewall, SIEM, cyber insurance, trained staff) bahut mahangi hai. Chhoti companies aur schools afford nahi kar paati — isliye wo aasan target ban jati hain.

12. Difficult Prosecution (Saza Dilana Mushkil)

Sahi saboot, sahi jurisdiction aur international cooperation ke bina **court me conviction rate bahut kam** hota hai. Criminal pakda bhi jaye to punish karna mushkil hai.

4.5 Diagram – Challenge vs Reason vs Solution

Exam me likhne ke liye ye compact table sabse useful hai:

Challenge	Problem Kyu Hai	Kya Solution Ho Sakta Hai
Anonymity	Criminal ki pehchan chhupi rehti hai	Digital footprint tracking, cyber forensics
Jurisdiction	Crime ka desh clear nahi	International treaties, mutual legal assistance
Borderless nature	Koi border nahi	Global cyber security cooperation
Evidence	Digital data volatile / deletable	Forensic imaging, proper chain of custody
Fast attacks	Seconds me ho jate hain	Automated detection, AI monitoring
Encryption	Data lock ho jata hai	Legal decryption procedures, key escrow
Technology change	Laws piche reh jate hain	Regular law updates
Lack of awareness	User khud dhoka kha jata hai	Cyber awareness campaigns, education
Low conviction	Proof + jurisdiction mushkil	Stronger laws, cyber courts

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):
1. Challenge = cyber crime ko rokne / pakadne me aane wali problem.
2. Sabse bada challenge = **Jurisdiction + Anonymity** (crime ka desh aur criminal ki pehchan).
3. Evidence digital hota hai — mit sakta hai, isliye jaldi collect karna zaroori.

Viva Q1. Cyber crime ko pakadna itna mushkil kyu hai?
Ans: Criminal chhupa hota hai (VPN/fake ID), crime country se bahar hota hai (jurisdiction), aur digital evidence mit jata hai.

Viva Q2. Insaan sabse bada weak point kyu?
Ans: Kyonki user OTP/share karta hai, unknown links par click karta hai aur weak passwords rakhta hai.

MEMORY TRICK

Challenge = Chhup-kar kaam karne wala criminal. Jaise chor raat ke andhere me chhup jata hai, waise cyber criminal VPN ke andhere me chhup jata hai — pakadna mushkil isliye kyunki koi face hi nahi dikhta.

PART 5 – CLASSIFICATIONS OF CYBER CRIMES

Unit 1 ka sabse important chapter. Isme hum syllabus ke 4 classifications ko ek-ek karke detail me padhenge: **Email Spoofing, Spamming, Internet Time Theft, Salami Attack**. Har ek me pattern same rahega — Example → Definition → Diagram/Flowchart → Prevention → Exam Notes.

5.0 Overview – Cyber Crimes Ki Classification

FLOWCHART

```
Classifications of Cyber Crimes
          │
 ┌────────┬────────┬────────┬─────────┐
 │                  │                  │                  │
Email              Spamming           Internet           Salami
Spoofing                              Time Theft         Attack
```

Iske alawa bhi kai classifications hain (Phishing, Hacking, Cyber Stalking, Software Piracy, etc.) — lekin syllabus me yahi 4 main hain, aur inhi ko exam me detail me likhna hai.

5.1 EMAIL SPOOFING

PRACTICAL EXAMPLE – REAL LIFE

Riya ko ek email mila jo **bilkul uske bank se dikh raha tha** — sender: **support@sbibank.com**.
Email me likha tha: *"Aapka account block hone wala hai. Verify karne ke liye is link par click karein."*
Riya ne link par click kiya — ek page khula jo bank ke page jaisa hi tha. Usne apni user ID, password aur OTP daal diye.
Agle hi minute uske account se **Rs. 40,000** nikal gaye.

Sach: wo email bank ne nahi bheja tha. Sender ka address **fake** tha — attacker ne bank jaisa dikhne ke liye bana diya tha. **Isi ko Email Spoofing kehte hain.**

DEFINITION

Email Spoofing ek cyber crime hai jisme attacker email ke **sender address ko fake** karke aisa dikhata hai jaise email kisi trusted person / company se aaya ho — taaki victim dhoka kha kar apni sensitive information (password, OTP, bank details) de de ya malware install kar le.

Simple: Jisne bheja hai uska naam jhootha likhna, taaki tumhe lagge ki trusted logon ne bheja hai.

Kaise Kaam Karta Hai (Diagram)

DIAGRAM – HOW EMAIL SPOOFING WORKS

```
Attacker
    │  creates a fake sender address
    ▼
Fake Email  (From: support@sbibank.com)
    │  sent over the internet
    │  (SMTP protocol sender ko verify nahi karta)
    ▼
Victim ka Inbox  (email trusted dikhta hai)
    │  victim clicks the link / replies
    ▼
Password / OTP chori  OR  Malware install
```

IMPORTANT POINTS

Ye kyun possible hai? Email bhejne wala protocol **SMTP (Simple Mail Transfer Protocol)** sender ka address **verify nahi karta**. Jaise koi letter par bina pakke hue jhootha naam likh de — post office rokta nahi. Waise hi email system bhi sender ko check nahi karta.

Types of Email Spoofing

Type	Kya hota hai	Example
1. Display Name Spoofing	Sirf naam trusted dikhata hai, email ID alag hoti hai	Riya Verma <riya123@gmail.com>
2. Legitimate Domain Spoofing	Same company ka domain hack karke use karta hai	support@sbibank.com (hacked)
3. Look-alike Domain Spoofing	Similar dikhne wala domain	sbibank.com vs sbibank-secure.com
4. Reply-to Spoofing	Reply jab karo to attacker ke paas jata hai	Reply-to: fraud@fake.com

Email Spoofing vs Phishing

Spoofing = sender address fake karna (technique). Phishing = dhoka dena (purpose). Spoofing ka istemal phishing karne ke liye hota hai. Matlab — spoofing chabi hai, phishing taala hai jo wo khulta hai. Dono aksar saath saath aate hain.

Prevention (Bachao Ke Tarike)

- Sender ka email address **dhyan se dekho** (domain check karo).
- Link par **hover** karke URL check karo — click karne se pehle.
- Urgent / dhamki wale messages par suspicious raho.
- **OTP kabhi kisi ko na do** — bank bhi OTP nahi maangta.
- Company/bank par call karke verify karo.
- Organizations apne email par **SPF, DKIM, DMARC** records enable karein.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):
1. Email Spoofing = sender ka address fake karna.
2. Target = trust jeet kar password/OTP churana.
3. Ye isliye possible hai kyunki **SMTP sender ko verify nahi karta**.

Viva Q1. Email spoofing me kya fake hota hai?
Ans: Sender ka address / identity.

Viva Q2. Spoofing aur phishing me kya relation hai?
Ans: Spoofing ek technique hai jo phishing (dhoka) karne ke liye use hoti hai.

5.2 SPAMMING

PRACTICAL EXAMPLE – REAL LIFE

Aman ka mobile har din **30 se zyada spam messages** se bhara rehta hai:
• “Aapko sirf Rs. 99 me smartphone milega!”
• “Congratulations! Prize jitne ke liye reply karein”
• Random loan / gambling / casino ads
Aur uske Gmail me bhi 100 unread promotional emails — asli emails dhoondna mushkil ho jata hai.
Agar Aman inme se kisi link par click kar de — malware ya fraud. **Yehi Spamming hai.**

DEFINITION

Spamming ek cyber crime hai jisme kisi ko **bina uski permission ke bulk (bahut zyada) unwanted messages, emails ya advertisements** bheje jaate hain.

Simple: Kisi aur ka inbox/mobile bina pooche-jhadhe ads aur bekaar messages se bhar dena.

FLOWCHART

```
Spammer
   │  collects many email / phone lists
   ▼
Sends bulk unwanted messages
   │
   ├──────────────────────┐
   │                            │
   ▼                            ▼
Victim ka inbox/phone flooded   Links click hone par
time & bandwidth waste          virus / fraud
```

Types of Spam

Type	Kya hai
Email Spam	Bulk promotional / junk emails
SMS Spam	Bulk messages on mobile
Comment Spam	Blogs / forums par fake comments (link ke saath)
Search Spam	Search results me fake pages dikhana
Social Media Spam	Fake accounts se bulk messages / tags

Spamming Ka Nuksan (Harm)

- **Time waste** — real messages dhoondne me time lagta hai.
- **Bandwidth / data khatam** — mobile data aur internet bill badta hai.
- **Real email miss** ho jate hain (important mail spam me dab jati hai).
- **Phishing aur malware ka zariya** — spam ke andar harmful links hote hain.
- **Fraud ka gate** — spam me fake offers lottery scams bheje jate hain.

Spam vs Scam

Spam = koi bhi unwanted bulk message. **Scam** = us message ke andar ka dhoka (paisa chori ki koshish). Spam scam ka **carrier** ban sakta hai — spam message ke andar scam ka link hota hai.

Prevention (Bachao Ke Tarike)

- **Spam filter** ON rakho (email apps me default hota hai).
- Apna email / phone number **public jagah share na karo** (social media, random sites).
- Trusted sites par hi **unsubscribe** karo.
- **Unknown links par click mat karo** — reply bhi mat karo.
- Spam ko **report** karo.
- Sign-ups ke liye **alag email ID** banao (junk email).

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):
1. Spamming = bina permission ke bulk unwanted messages bhejna.
2. Spam ka sabse bada nuksan = time/bandwidth waste + phishing ka zariya.
3. Spam vs Scam = spam carrier hai, scam dhoka hai.

Viva Q1. Spamming ko cyber crime kyu mana jata hai?
Ans: Kyunki ye bina permission ke doosron ke inbox/data par bojh daalta hai aur fraud ka zariya banta hai.

Viva Q2. Spam se bachne ka best tarika?
Ans: Spam filter, email public na karna, unknown links par click na karna.

5.3 INTERNET TIME THEFT

PRACTICAL EXAMPLE – REAL LIFE

Ek company har month internet bill deti hai — Rs. 50,000. Employees ka kaam 6 ghante ka hai. Lekin ek employee **Suresh** roz 3 ghante **office internet par online movies dekhta hai, games download karta hai, file sharing karta hai**.
Iska asar:
• Office ka internet slow ho jata hai — baaki employees ka kaam rukta hai.
• Company ka bandwidth aur paisa bekaar jaata hai.
• Company ke data par risk aata hai (pirated content download).
Yahan company ka **paid internet time aur bandwidth bina permission ke personal kaam me** use hua — **Internet Time Theft**.

PRACTICAL EXAMPLE – REAL LIFE

Kabir ke neighbour ne Kabir ka **Wi-Fi password guess** kar liya. Wo roz Kabir ke internet se downloads karta hai. Ab:
• Kabir ka internet slow ho jata hai.
• Data / FUP khatam ho jata hai, bill badhta hai.
Sabse khatarnak baat — agar neighbour ne internet par koi **galat kaam** kiya (illegal download, fake post), to IP address ke hisaab se **police Kabir ke ghar aayegi** kyunki login Kabir ke connection se hua tha. Kabir ko apni bezati ke saath apni bezosai sabit karni padegi.

DEFINITION

Internet Time Theft ek cyber crime hai jisme koi person kisi doosre ki **paid internet connection, bandwidth, ya kaam ka samay bina permission ke apne personal / illegal kaam me use** karta hai.

Simple: Doosre ka internet / doosre ka kaam ka time — bina pooche apne kaam me lagaana.

FLOWCHART

```
Internet Time Theft
      │
┌──────────────┬──────────────┐
│                             │
Type A: Employee Time Theft      Type B: Connection Theft
      │                             │
Office internet / working        Kisi aur ka Wi-Fi / data
hours → personal browsing       → bina permission use
      │                             │
Company ko loss:                 Victim ko loss:
salary + bandwidth +           bill, slow internet,
productivity                    illegal activity ka blame
```

Examples

- Office me duty hours par **online gaming / movie** dekhna.
- Neighbour ka **Wi-Fi password hack / guess** karke use karna.
- Kisi ka **SIM data balance** bina permission use karna.
- Shared connection (network spoofing) bina permission use karna.

Prevention (Bachao Ke Tarike)

- **Strong Wi-Fi password + WPA2/WPA3** encryption.
- Company me **internet usage policy aur monitoring**.
- **MAC address filtering** — sirf known devices hi connect hon.
- **Guest network** — visitors ke liye alag network.
- Default password turant **change** karo, time-to-time password change karo.
- Office kaam aur personal kaam ke liye **alag connections**.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):
1. Internet Time Theft = doosre ka paid internet/bandwidth/time bina permission use karna.
2. Do types = Employee Time Theft aur Connection Theft.
3. Danger = illegal activity ka IP blame victim par aata hai.

Viva Q1. Internet Time Theft me kaun kaun loss me hota hai?
Ans: Company (salary/bandwidth/productivity) ya victim (bill/slow internet/blame).

Viva Q2. Isse bachne ke liye sabse zaroori kya hai?
Ans: Strong password + WPA2/WPA3 aur usage monitoring.

5.4 SALAMI ATTACK / SALAMI TECHNIQUE

PRACTICAL EXAMPLE – REAL LIFE

Ek bank me **10 lakh logon ke accounts** hain. Har account par interest credit hota hai.
Interest nikalte waqt exact value aati hai, jaise: Rs. 100.0047, Rs. 55.2358 ...
Bank aam taur par **paise tak round** karta hai: Rs. 100.0047 → Rs. 100.00 credit.

Attacker (employee ya hacker) system me **chhota sa code** daal deta hai jo round karne ke baad bacha fraction **(0.0047, 0.2358) alag account me transfer** kar deta hai.
• 1 account se kuch nahi — sirf Rs. 0.0047.
• 10 lakh accounts se — 0.0047 × 10,00,000 = Rs. 4,700 chupchaap.
• Agar fraction 0.50 se 0.90 ka ho to — **lakhon rupaye**.

Kisi bhi account holder ko notice nahi hota — kyunki usko hamesha puri value hi mil rahi thi (round off). Total bekaar me kai lakh rupaye chori — **ye Salami Attack hai.**

DEFINITION

Salami Attack (Salami Technique) ek cyber crime hai jisme bahut se accounts / files se **bahut chhote-chhote amounts** is tarah cut kiye jaate hain ki **koi notice na kare**, lekin **total badi amount** ban jati hai.

Simple: Har jagah se thoda-thoda — kisi ko pata na chale — lekin total milakar bada gola.

Naam 'Salami' Kyu?

Salami (sausage) ko **bahut patle-patle slices** me kaata jata hai — har slice kisi ko nazar nahi aata, lekin dheere-dheere **poori sausage khatam** ho jaati hai. Bilkul isi tarah is attack me **chhote-chhote hisse** kaate jaate hain — isliye naam **Salami Attack**.

FLOWCHART

```
Bank system (lakhon accounts)
      │
      ▼
Interest rounding: Rs 100.0047 → 100.00
      │
      ▼
Fraction (0.0047) attacker ke account me
      │
      ▼
Har account se chhota loss (notice nahi hota)
      │
      ▼
Lakhon accounts × chhota amount = Badi chori
      │
      ▼
Kisi ko pata nahi chalta / trace karna mushkil
```

Formula (Diagram)

DIAGRAM

```
Chhota Amount  ×  Bahut Saare Accounts  =  Badi Chori

Rs. 0.0047   ×   10,00,000 accounts    =  Rs. 4,700
Rs. 0.50     ×   10,000,000 accounts   =  Rs. 50,00,000

(Har account ko loss itna chhota ki wo notice hi na kare)
```

Salami Technique Ke Aur Examples

- **Billing system** — har bill par Rs. 1 extra (rounding), 1 lakh bills = Rs. 1 lakh chori.
- **Currency exchange** — fraction ka paisa (paisa value) wapas na dena, collect karte jaana.
- **Payroll / pension** — har employee ke salary se Rs. 2-3 katna — kisi ko pata nahi chalta.

Prevention (Bachao Ke Tarike)

- **Audit trails** — har transaction ka pura record.
- Rounding / fraction ke changes par **monitoring**.
- **Internal controls + dual approval** — ek banda akela change na kar sake.
- **Regular reconciliation** — saare accounts match karna (deposits vs records).
- Fraction data ko bhi **log** karna.
- **Whistle-blower policy** — employees bina dar ke report karein.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Salami Attack = bahut chhote-chhote amounts, bahut saare accounts se, notice na hote hue — total bada.
2. Naam 'Salami' isliye — jaise sausage ke patle slices kaatte hain, waise hi chhote hisse kaate jaate hain.
3. Formula: Chhota amount × Lakhon accounts = Badi chori.
4. Ye itna dangerous hai kyunki **detection bahut mushkil** hota hai — koi bhi ek account owner notice nahi karta.

Viva Q1. Salami attack me har account ko kitna loss hota hai?
Ans: Bahut chhota (paisa-fraction) — isliye notice nahi hota.

Viva Q2. Salami attack ko pakadna mushkil kyu?
Ans: Kyunki har jagah amount na ke barabar hai, sirf total bada hota hai — aur rounded values ki wajah se kisi ko difference dikhta nahi.

MEMORY TRICK

Salami = Sausage ki patli-slice. Jaise patle-patle slices se poori sausage khatam — waise hi chhote-chhote cut se poori daulat khatam. Socho: **1 rupaya × 1 crore accounts = 1 crore rupaye** — kisi ko pata bhi nahi chala.

APPENDIX D – QUICK REVISION SHEET (UNIT 1)

Sabse Pehle Yaad Rakho — One-Line Definitions

Topic	Ek Line Me
Cyber Security	Computers, networks, data ko attacks se protect karne ka process
Cyber Attack	System/data ko access, steal, damage karne ka intentional attempt
Cyber Crime	Computer/internet ke through ki gayi criminal activity
Email Spoofing	Sender ka address fake karke trusted dikhana
Spamming	Bina permission ke bulk unwanted messages bhejna
Internet Time Theft	Doosre ka paid internet/bandwidth/time bina permission use karna
Salami Attack	Chhote-chhote amounts, bahut saare accounts se, notice na hote hue

Memory Tricks

- Cyber Security = **Digital Ghar ki Security**.
- Cyber Attack = **Digital Chor**.
- Cyber Crime = **Digital Daku**.
- Salami = **Sausage ki patli-slice** — 1 rupaya × 1 crore accounts = 1 crore rupaye.
- CIA = **C**onfidentiality, **I**ntegrity, **A**vailability.
- Active Attack = **Badlaav karta hai**, Passive Attack = **Sirf dekhta hai**.

Rapid Fire Viva (10 Questions)

Q	Answer
Cyber security kya protect karti hai?	Computer, network, server, mobile, data
Cyber attack ka sabse common motive?	Paisa (financial gain)
Cyber crime ke 5 elements?	Offender, Tool, Victim, Act, Loss
Cyber crime ki 3 categories?	Against Person, Against Property, Against Government
Email spoofing me kya fake hota hai?	Sender ka address
Email spoofing kyu possible hai?	SMTP sender ko verify nahi karta
Spam ka carrier kya hota hai?	Scam / phishing / malware
Internet Time Theft ke 2 types?	Employee Time Theft, Connection Theft
Salami attack me kitna loss per account?	Na ke barabar (chhota) — isliye notice nahi hota
Sabse mushkil cyber crime pakadna?	Salami attack (koi notice nahi karta)

Note: Ye Unit 1 ka complete notes hai. Aage ke chapters (Active/Passive Attack detail, Malware types, Phishing, Virus etc.) same pattern par alag book me banenge.